WEEK 1 · SESSION 3

Enterprise Data Engineering, Analytics & AI

SQL & Data Engineering Concepts

Data Types · Modern Storage Systems · SQL Foundations

```
SELECT * FROM
  data_warehouse
WHERE layer = 'gold'

CREATE TABLE facts (
  id INT PRIMARY KEY,
  metric DECIMAL(18,2)
);
```

Duration: 1 Hour | Trainer: Prachi Kabra | Foundation Phase

What You'll Walk Away With

By the end of this session, you will be able to:

Read & Write SQL Data Types

Know INT, VARCHAR, DECIMAL, DATE, BOOLEAN, JSON — and when to use each one

Identify Modern Storage Systems

Distinguish Database, Data Warehouse, Data Lake, Lakehouse and Datamart with confidence

Understand the Data Engineer's Role

Know where data engineers sit in the modern data stack and what they actually build

Map Storage to Use Case

Match the right storage system to the right business problem — not just memorise definitions

Connect to the Bigger Picture

See how SQL, storage systems and data engineering link to what's coming in Weeks 2–16

Why Data Types Matter

A database that doesn't know what kind of data it holds is a liability, not an asset

Financial Miscalculation

A bank stores a loan amount as TEXT instead of DECIMAL. Arithmetic operations fail silently. A ₹50,000 loan appears as ₹5,00,000 in a report.

Date Logic Collapse

A hospital stores dates as VARCHAR. Sorting patients by admission date returns '9th Jan' after '10th Jan'. Triage decisions are made on wrong order.

Security Vulnerability

A login system stores passwords and usernames in the same VARCHAR column with no length constraint. A 2MB string crashes the auth service.

🔑 **The right data type enforces correctness at the storage level — before any code runs.**

SQL Data Types

Every column in every table has a type — choosing the right one is the first design decision a data engineer makes

🔢 NUMERIC

`INT / BIGINT`	Whole numbers	*Employee count, Order ID, Age*
`DECIMAL(p,s)`	Exact precision numbers	*Price: DECIMAL(10,2) = ₹9,99,999.99*
`FLOAT / REAL`	Approx. decimal	*Scientific measurements (use carefully)*
`SMALLINT /`	Small whole	*Rating (1–5), Status flags*

📝 TEXT / STRING

`VARCHAR(n)`	Variable length text	*Name VARCHAR(100), Email VARCHAR(255)*
`CHAR(n)`	Fixed length text	*Country code CHAR(2): 'IN', 'US'*
`NVARCHAR(n)`	Unicode / multilingual	*Hindi names, Arabic text, Emoji*
`TEXT / NTEXT`	Large text blocks	*Product descriptions,*

📅 DATE & TIME

`DATE`	Date only	*2024-01-15 — DOB, Order date*
`DATETIME`	Date + time	*2024-01-15 14:30:00 — Transactions*
`TIMESTAMP`	Auto system time	*Record created/updated tracking*
`TIME`	Time only	*Shift start: 09:00:00*

🔵 OTHER ESSENTIALS

`BIT / BOOLEAN`	True/False flags	*is_active=1, is_deleted=0*
`JSON`	Semi-structured data	*API payloads, config objects*
`UNIQUEIDENTIFIER`	UUID / GUID	*Global unique keys across systems*
`BINARY / BLOB`	Raw binary data	*Images, PDFs, encrypted data*

Data Types in Action

Designing a real table — every column choice is deliberate

```sql
CREATE TABLE Customers (

    CustomerID      INT              PRIMARY KEY,

    FullName        NVARCHAR(150)    NOT NULL,

    Email           VARCHAR(255)     UNIQUE NOT NULL,

    PhoneNumber     VARCHAR(15)      NULL,

    DateOfBirth     DATE             NOT NULL,

    CreditLimit     DECIMAL(12,2)    DEFAULT 0.00,

    IsActive        BIT              DEFAULT 1,

    CreatedAt       DATETIME         DEFAULT GETDATE()

);
```

💬 *Ask: 'What would break if PhoneNumber was stored as INT?'*

Why these choices?

INT

IDs are whole numbers — no decimals ever needed

NVARCHAR

Customer names may contain Unicode (Hindi, Arabic)

DECIMAL(12,2)

Credit limits need exact precision — FLOAT would round

BIT

Active/Inactive flag — only 2 states, 1 bit of storage

DATETIME

Record creation timestamp — auto-filled by database

The Modern Data Storage Stack

Four systems — different purposes, different strengths. A mature enterprise uses all four.

Database

OLTP — Online Transaction Processing

Real-time read/write operations for applications

`SQL Server · MySQL · PostgreSQL · Oracle`

⚡ Millisecond reads/writes

Best for: Orders, accounts, logins, inventory

Data Warehouse

OLAP — Online Analytical Processing

Historical analytics on structured, cleaned data

`Snowflake · Azure Synapse · BigQuery · Redshift`

📊 Seconds to minutes for complex queries

Best for: Monthly reports, KPIs, business intelligence

Data Lake

Raw Storage — All Formats

Store everything raw — structured, semi, unstructured

`Azure ADLS Gen2 · AWS S3 · Google GCS`

🌊 Low cost, high volume — not for fast queries

Best for: IoT dumps, logs, JSON, images, ML training data

Lakehouse

OLTP + OLAP — Best of Both

Combines Data Lake scale with Warehouse performance

`Databricks Delta Lake · Microsoft Fabric · Apache Iceberg`

🚀 Fast + scalable + ACID compliant

Best for: Streaming + batch, SQL + ML in one platform

💬 *'If you had to store every UPI transaction in India — 10 billion per day — which system do you start with?'*

Datamart — The Focused Layer

A Datamart is a subject-focused slice of a Warehouse — built for one team or one business question

Analogy: A Data Warehouse is a supermarket with everything. A Datamart is the dairy aisle — same building, curated for one specific need.

Sales Datamart	Finance Datamart	HR Datamart	Marketing Datamart
Revenue by region	P&L summary	Headcount trends	Campaign ROI
Lead conversion	Cost per department	Attrition rate	Click-through rates
Pipeline value	Cash flow	Hiring velocity	Customer segments
Top customers	Budget vs Actual	Salary bands	Ad spend

📐 Warehouse → Datamart flow: Central Warehouse stores all data → Datamarts pull department-specific subsets for faster, focused queries

Which System — When?

The most important skill: matching the right storage system to the right business need

Dimension	Database	Data Warehouse	Data Lake	Lakehouse	Datamart
Primary Use	App transactions	Analytics/BI	Raw storage	Analytics+ML	Dept reporting
Data Format	Structured only	Structured	Any format	Any format	Structured
Query Speed	⚡ Very fast	🔵 Fast	🐢 Slow	🚀 Fast	⚡ Fast
Data State	Live/current	Historical	Raw/unprocessed	Processed	Curated subset
Who uses it	Dev / App	Analyst / BI	Data Eng / Sci	All roles	Business users
Cost	Medium	High	Very Low	Medium	Low
Example tool	SQL Server	Snowflake	Azure ADLS	Databricks	Power BI dataset

💡 *In Week 2 we model data for the Warehouse. In Week 12–13 we build the Lakehouse on Databricks.*

⚡ QUIZ TIME

3 Questions · 4 Minutes · No conferring yet!

Q1 → A column stores product prices like ₹1,499.99. Which data type should you use — and why NOT use FLOAT?

Think: precision vs. approximation

Q2 → Your company runs 10 million daily sales transactions AND needs monthly revenue reports. Name the TWO storage systems you need and why.

Think: different jobs, different tools

Q3 → What is a Datamart, and how is it different from a Data Warehouse?

Think: scale vs. focus

Where the Data Engineer Lives

Understanding the full stack — and your place in it

Source Systems
ERP, CRM, APIs, IoT

→

Ingestion Layer
Pipelines, Batch, Stream
YOU BUILD THIS

→

Storage Layer
Lake → Warehouse → Datamart
YOU BUILD THIS

→

Transform & Model
SQL, dbt, Spark, Python
YOU BUILD THIS

→

Serve Layer
BI Tools, APIs, ML

→

Consumer
Analyst, Executive, App

What a Data Engineer actually builds:

- Pipelines that move data from source to lake
- SQL scripts that transform raw data into clean tables
- Models that shape data for analysts
- Jobs that schedule and automate data flows

Real World — Choose the Right System

Four scenarios — four different answers. Can you match them before reading the right column?

🏦 A bank needs to process 5 million credit card transactions per day — each one must succeed or fail atomically with no data loss.

→ **Relational Database (SQL Server / Oracle)**

OLTP: each transaction is a real-time write that needs ACID guarantees — rollback if anything fails.

🏭 A manufacturing company wants to analyse 3 years of production data to find the machines with the highest failure rates.

→ **Data Warehouse (Snowflake / Synapse)**

OLAP: historical aggregation across millions of rows — built for complex analytical queries, not live writes.

🤖 A data science team needs to train an ML model on 10TB of raw CCTV footage, IoT sensor logs and social media data.

→ **Data Lake (Azure ADLS / AWS S3)**

Unstructured + semi-structured at scale — store everything raw at low cost before any schema is enforced.

📊 The CFO wants a self-service dashboard showing only Finance KPIs — fast, governed and simple to maintain.

→ **Datamart (curated from Warehouse)**

Subject-focused subset of the warehouse — small, fast, governed for one business function.

Session Summary

1

Data types enforce correctness at storage level — wrong type = silent corruption or system failure

2

Key SQL types: INT/BIGINT for IDs, DECIMAL for money, NVARCHAR for text, DATE/DATETIME for time, BIT for flags, JSON for semi-structured data

3

Database (OLTP) — for live transactions. Data Warehouse (OLAP) — for historical analytics

4

Data Lake — raw storage of any format at scale. Lakehouse — combines Lake + Warehouse in one platform

5

Datamart — a department-focused subset of the warehouse, built for fast, governed, self-service reporting

6

A Data Engineer builds the pipelines, models and jobs that connect all these systems together

⏭ Next: Week 2 — Data Modelling Concepts (Fact tables, Dimension tables, Star & Snowflake schema, ER Diagrams)

Questions?

"A data engineer who understands storage systems is worth ten engineers who just write queries."

📝 Before Week 2

Think of one table you would create for a company you know.
Name 5 columns — and write the correct SQL data type for each one.
We'll review your designs at the start of Week 2, Session 1.